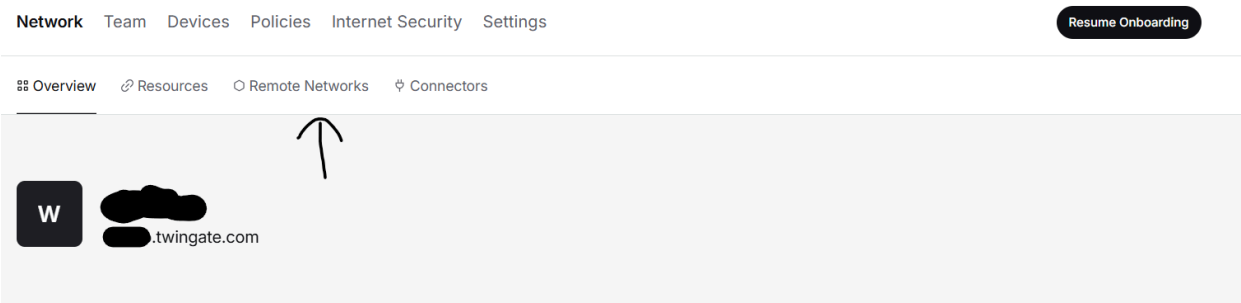
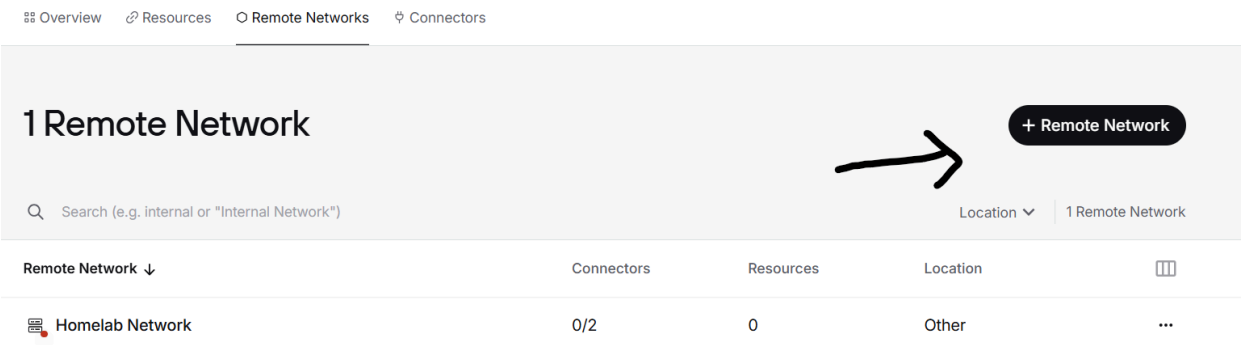
Setting Up your Twingate account

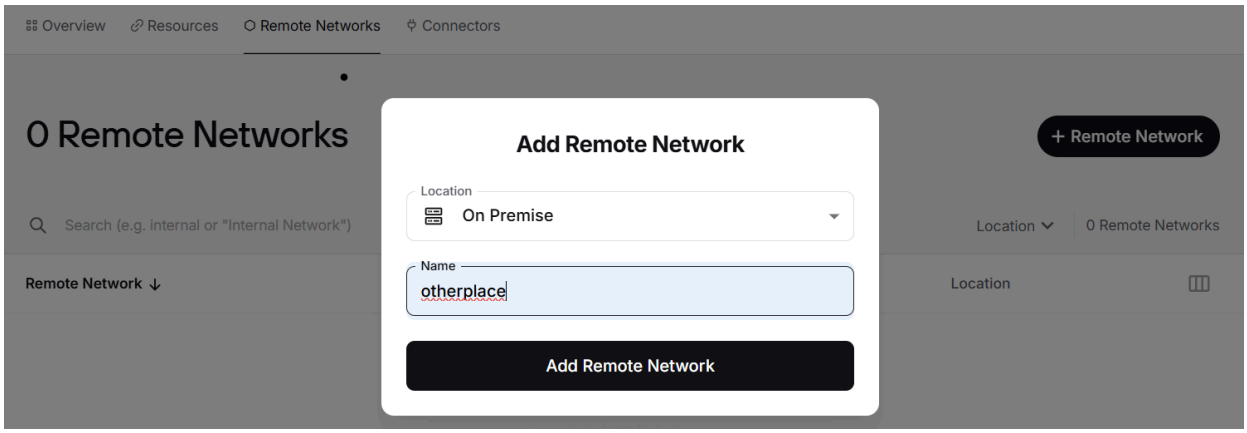
1. Creating your new Twingate Network
2. Use the gmail that you want to set Twingate up
3. Set any name for your network name
    a. **<any_name>**.twingate.com for your **Network URL**
4. Sign up with email your preferred
5. Click on **"Skip to Admin Console"**
6. Click on **Remote Networks**



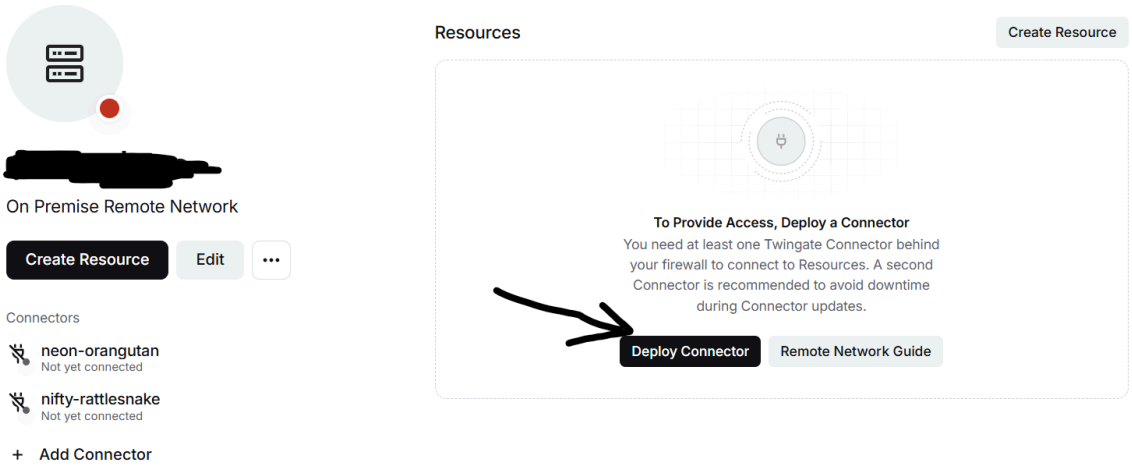
7. Click on "**Remote Network**"



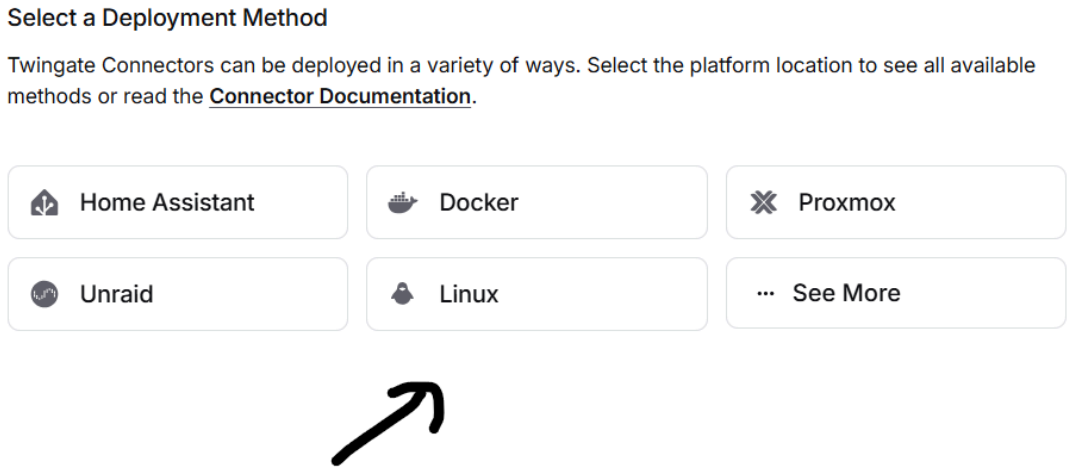
8. In the **Location** section, choose the **On Premise**



9. Click on the **Remote Network** that you just created
10. Before clicking the **Deploy Connector** option, you must set up the device that you want to remotely connect to, in this case I will be using Raspberry Pi 5 with Raspberry Pi OS installed
11. After setting up a device, now you can choose **Deploy Connector** option



12. Choose the OS that you have installed on the remote Device, for my case, I did installed Raspberry Pi OS



13. Click on **Generate Tokens**. After that, it will reload the website automatically and you have to sign in again to Twingate with the previous email you have

30000-31000. For UDP and QUIC for HTTP/3 Ports, you may limit to 1-65535.

Access all Resources in the Network

Ensure that Connectors have both permission and routing rules to access private Resources. Resources will be resolved and routed from the Connector host.

Deploy close to Resources

Minimize the amount of hops on the network.

Step 2. Generate Tokens

You'll need to re-authenticate to generate Connector tokens. These tokens will be used in the Linux commands below.

Don't reuse Connector tokens

Connector tokens are unique to each individual Connector and they cannot be shared between Connectors.

Access Token
Not yet generated
Copy Token

Refresh Token
Not yet generated
Copy Token

Generate Tokens

Step 3. Customize Linux Command

The Connector will be deployed as a `systemd` service. Learn more

Local network connection logs (optional)

Step 4. Install and Launch Connector Service

Run the following command in the shell (we support Ubuntu, Fedora, and CentOS) to install and launch the Connector.

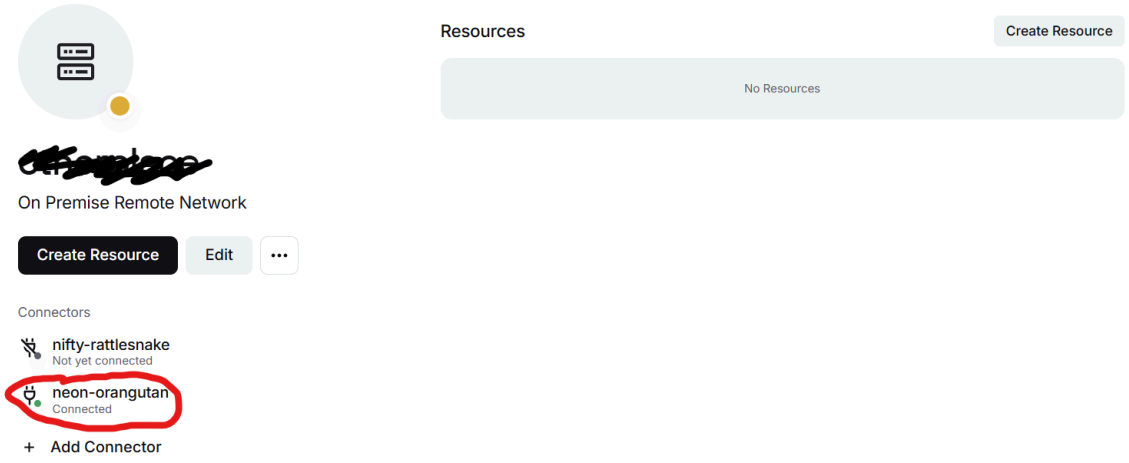
```
curl "https://binaries.twingate.com/connector/setup.sh" | sudo TWINGATE_ACCESS_TOKEN="[redacted]
```

Copy Command

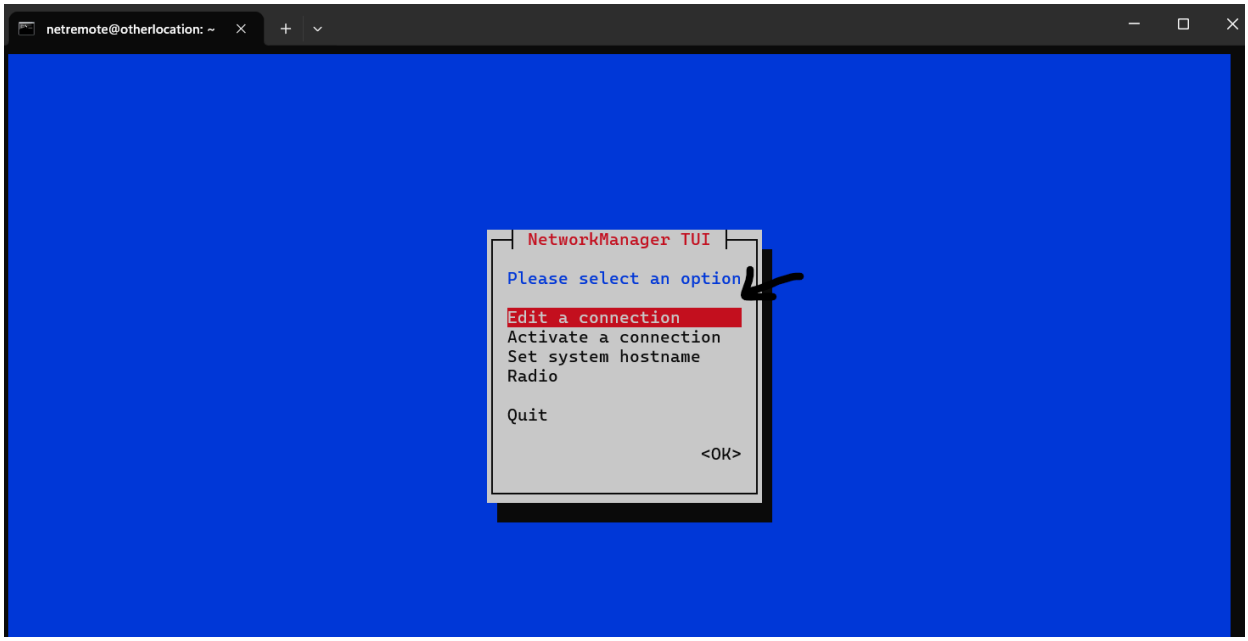
Step 5. Wait for Connection

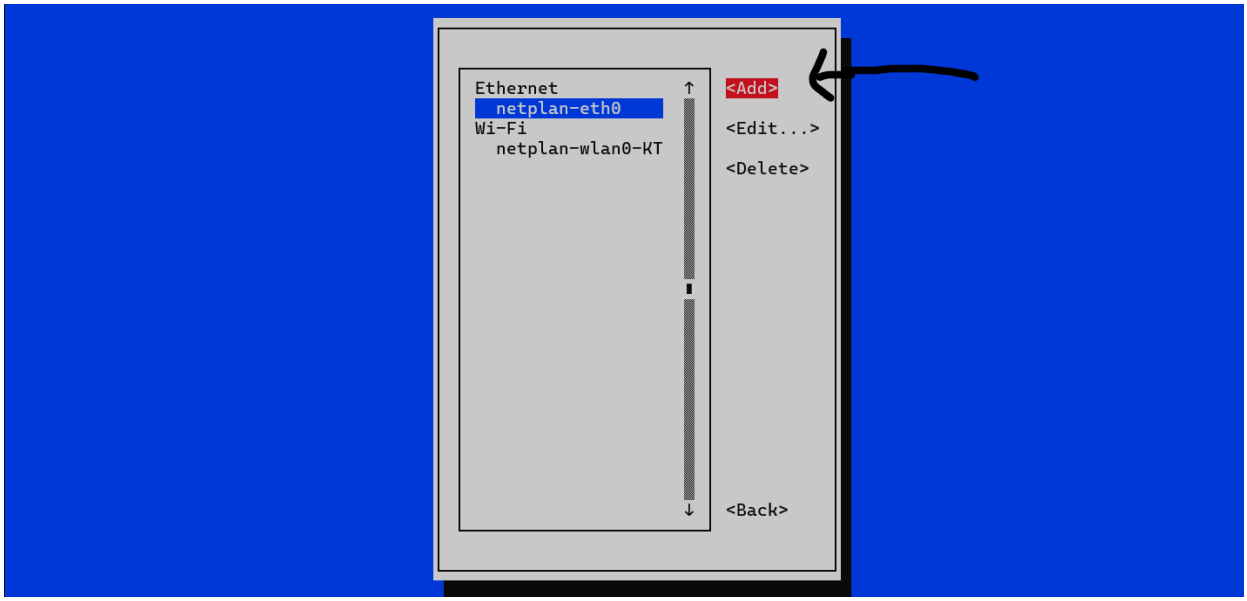
After running the deployment command, the Connector should automatically connect to Twingate. If you're still having trouble, double check that you're following our best practices. Learn more

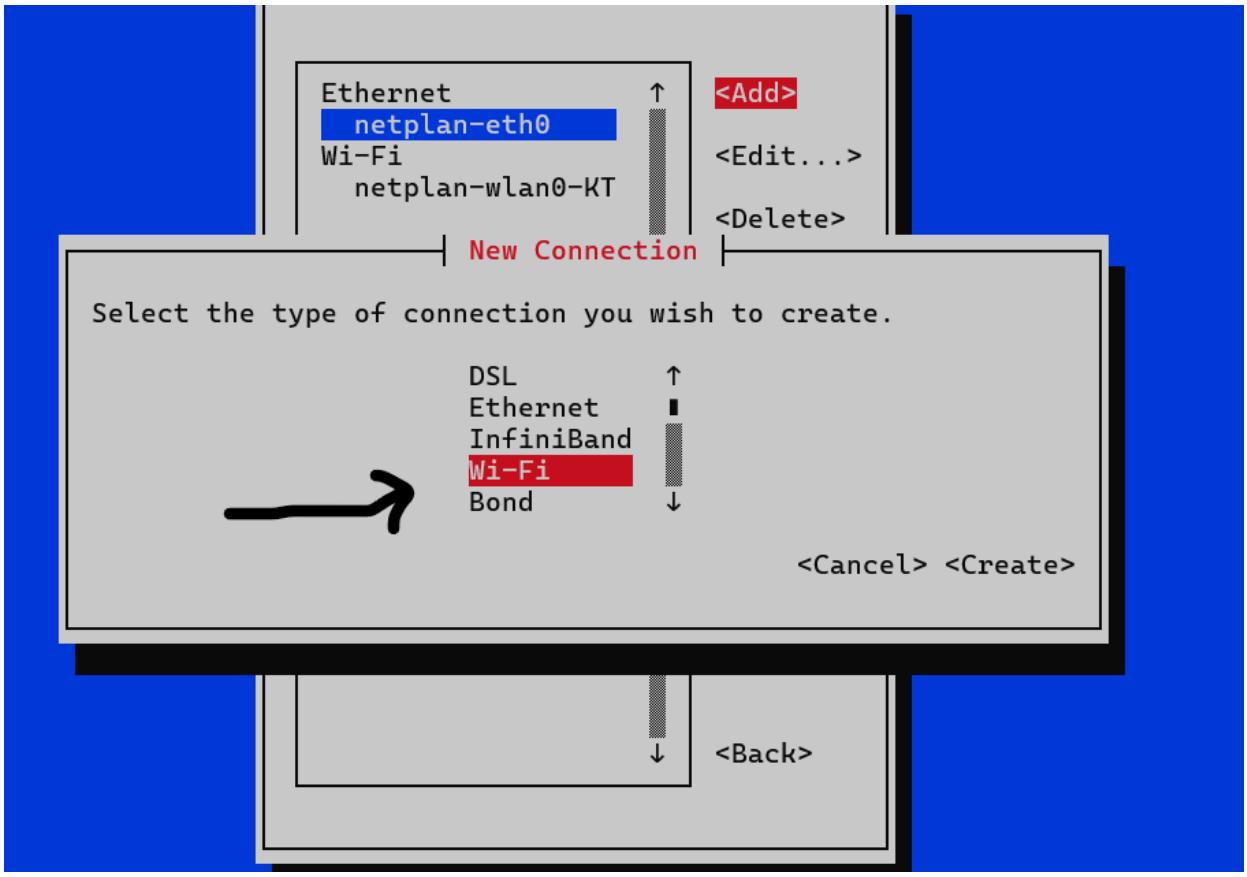
14. If you are on Linux OS, before you copy command in step 4, make sure to do
    a. Sudo apt update && sudo apt upgrade -y
    b. Then copy command in step 4 (in the image)
15. After this, the device will automatically connect to Twingate, and you can see as follow

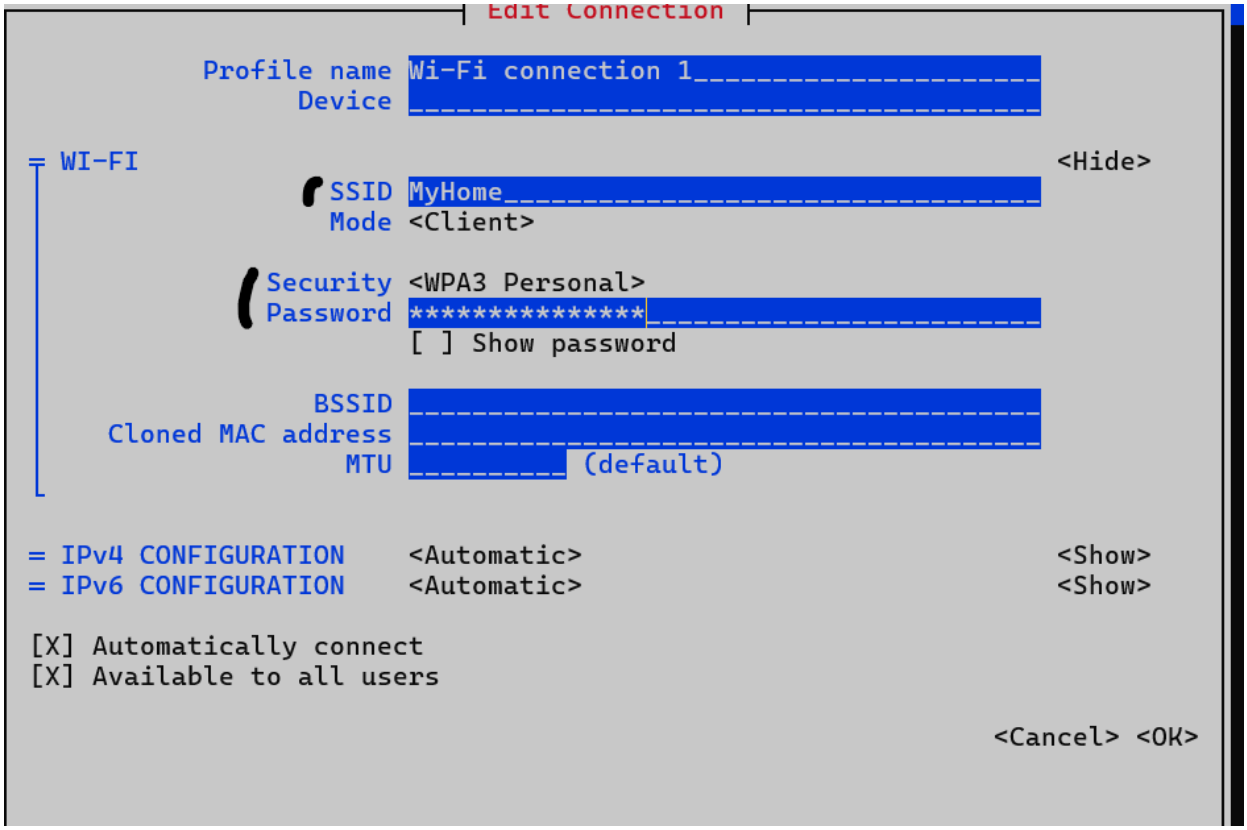


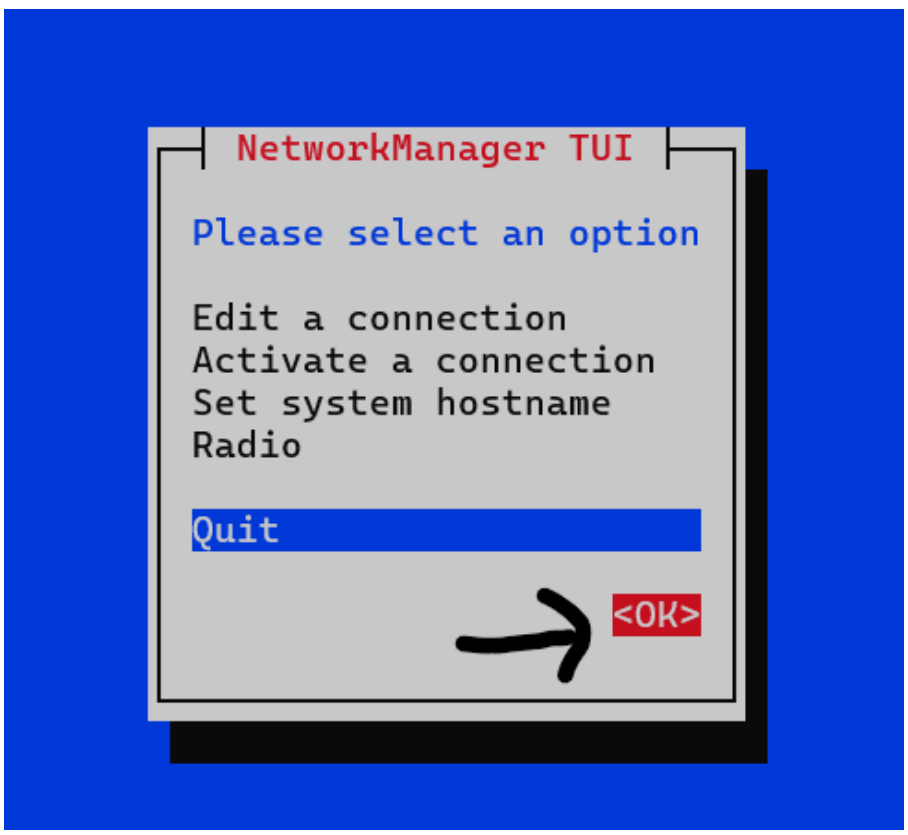
16. Before sending the remote device to other location, make sure to set up the other location's first
    a. Type command: sudo nmtui
    b. From then, follow these instructions


netremote@otherlocation: ~
NetworkManager TUI
Please select an option
Edit a connection
Activate a connection
Set system hostname
Radio
Quit
<OK>

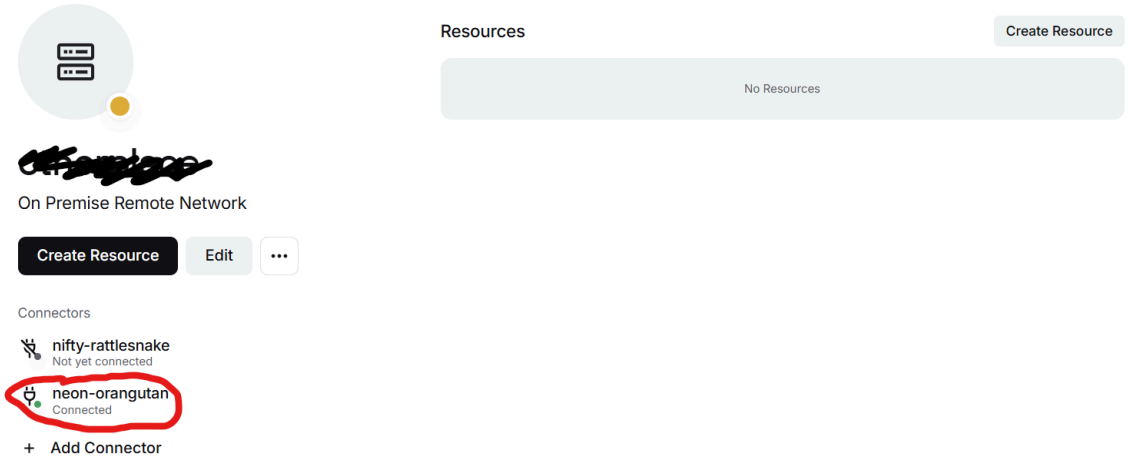

Ethernet
netplan-eth0
Wi-Fi
netplan-wlan0-KT
<Add>
<Edit...>
<Delete>
<Back>


Ethernet
netplan-eth0
Wi-Fi
netplan-wlan0-KT
<Add>
<Edit...>
<Delete>
New Connection
Select the type of connection you wish to create.
DSL
Ethernet
InfiniBand
Wi-Fi
Bond
<Cancel> <Create>
<Back>


Edit Connection
Profile name Wi-Fi connection 1
Device
= WI-FI
<Hide>
SSID MyHome
Mode <Client>
Security <WPA3 Personal>
Password **********
[ ] Show password
BSSID
Cloned MAC address
MTU (default)
= IPv4 CONFIGURATION <Automatic> <Show>
= IPv6 CONFIGURATION <Automatic> <Show>
[X] Automatically connect
[X] Available to all users
<Cancel> <OK>


NetworkManager TUI
Please select an option
Edit a connection
Activate a connection
Set system hostname
Radio
Quit
<OK>

17. Once you enter OK, now you can send the remote device to the location you want to remote to
18. Make sure when you open the Twingate again, the remote device still say connected on Twingate



OPTIONAL:

Now we have come to the option session, where you don't need to follow, but it can come in handy when you need to know more about the remote network, especially devices connected to that network, like IP address, and ports that are active on each device in that network.

In order to do these steps.

1. First you need to copy the API of your Twingate

Twingate

Network Team Devices Policies Internet Security Settings

Back to Remote Networks

General
Billing
Identity Providers
Device Integrations
Reports
API
Notification Channels
Notifications

General

Network

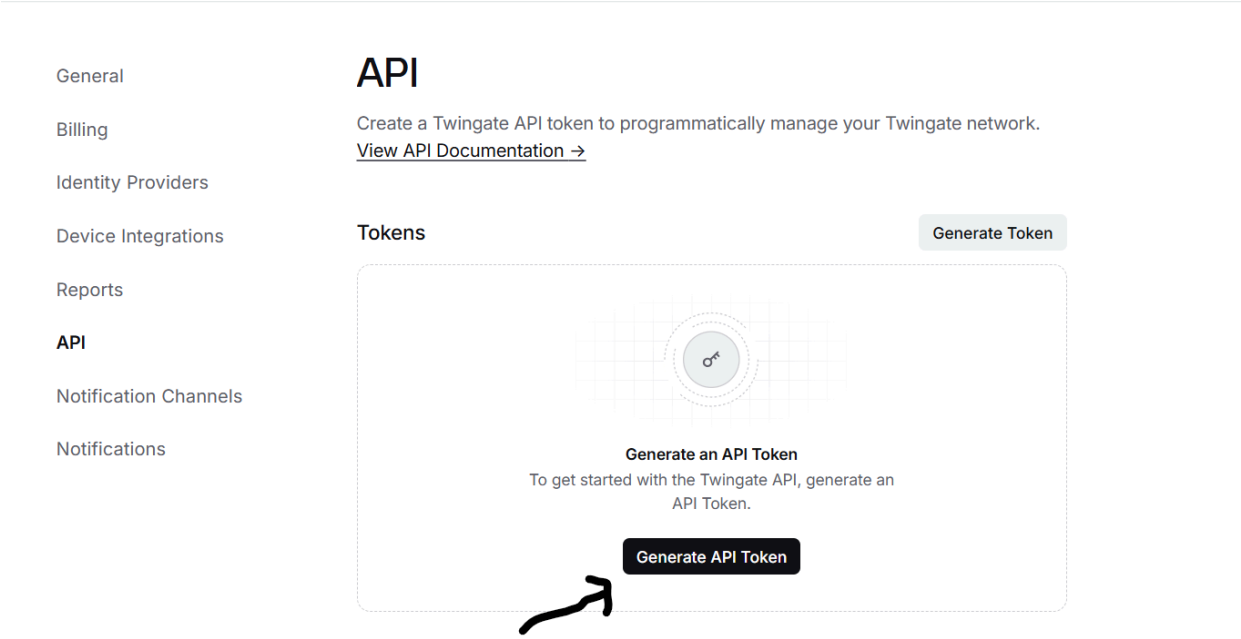
[redacted]

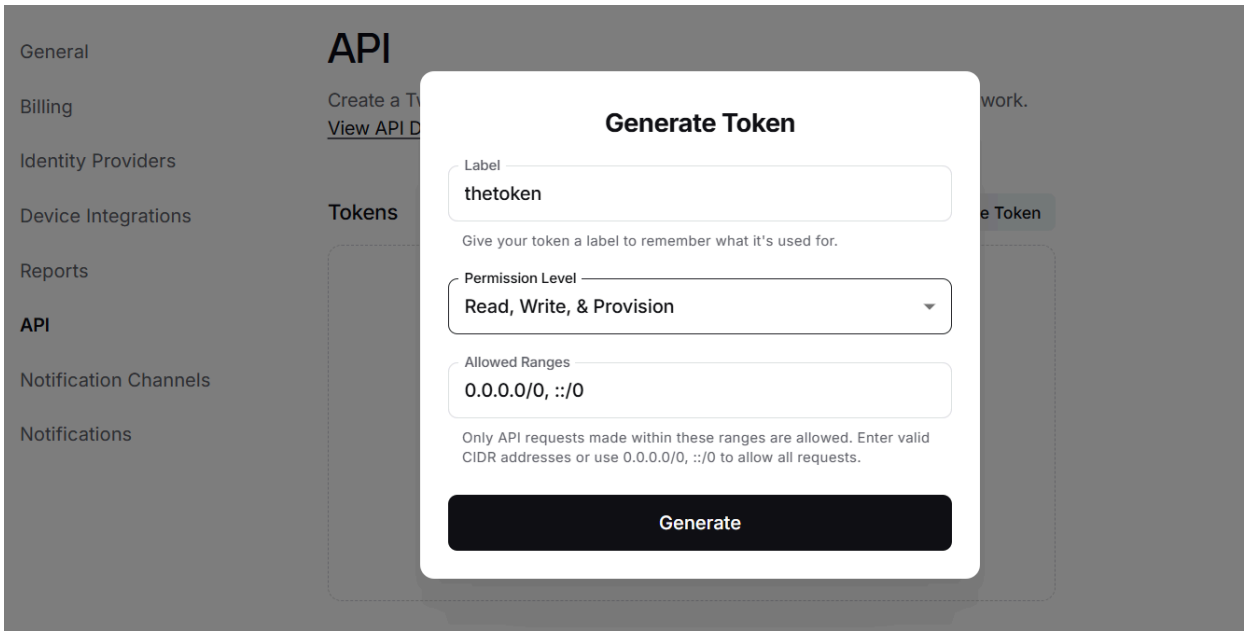
Admin Console Security

Authenticate every **1 hour**

MFA **not required**

Enforced when an admin signs in to the Admin Console.





2. Follow the steps as in the images in order, then a token will be generated and copy the Token in somewhere safe so that you can copy and paste in the remote device's terminal later on
3. Create the Credentials File: Credentials (which contains API's token) are stored in a separate .env file and never hardcoded in the script. This is a security best practice

a. nano secure_key.env
b. Inside the env file, copy and paste this
    i. TWINGATE_API_URL=https://**yoursubdomain**.twingate.com/api/graphql/
    ii. TWINGATE_API_KEY=your_api_key_here
    iii. TWINGATE_NETWORK_NAME=your_network_name

Notes: Change the **yoursubdomain** to match your network name, which is from **<your_name>,twingate,com**, copy the **API's token** in the TWINGATE_API_KEY and copy your **Remote Network's name** to TWINGATE_NETWORK_NAME

4. Create and activate virtual environment and install required package
    a. python3 -m venv twingate_env
    b. source twingate_env/bin/activate
    c. pip install "gql[requests]"
5. Next step, running a python script
    a. Before copying the python script, you need to create python file
        i. nano twingate_automate.py
        ii. Then copy and paste the python code from other session
6. Running the Script
    a. First you need to dry run the test first - previews what would be created without making changes. Always run this first.
        i. python3 twingate_automate.py --env-file secure_key.env --dry-run
        ii. Expected Dry-Run Output:
            1. [INFO] Connecting to Twingate API: https://yournetwork.twingate.com/api/graphql/
            2. [INFO] Target network: yournetwork
            3. [INFO] Searching for target networks...
            4. [INFO] Found network: yournetwork (ID: UmVtb3RlTmV0d29yazozMDIxODQ=)
            5. [INFO] --- DRY RUN MODE: no changes will be made ---
            6. [INFO] Connector: your-connector-name
            7. [INFO] [DRY RUN] Would create: your-connector-public-x-x-x-x (x.x.x.x)
            8. [INFO] [DRY RUN] Would create: your-connector-private-192-168-1-x (192.168.1.x)
    b. Then actually live run to actually create the resources in Twingate
        i. python3 twingate_automate.py --env-file secure_key.env
        ii. Expected Live Run Output:
            1. [INFO] Connector: your-connector-name
            2. [INFO] Created: your-connector-public-x-x-x-x | ID: UmVzb3VyY2U6... | Address: x.x.x.x

    3. [INFO] Created: your-connector-private-192-168-1-x | ID: UmVzb3VyY2U6... | Address: 192.168.1.x
    4. [INFO] Done. 2 resource(s) created, 0 skipped.
7. Verify resources appeared in Twingate Admin Console and assign to a group